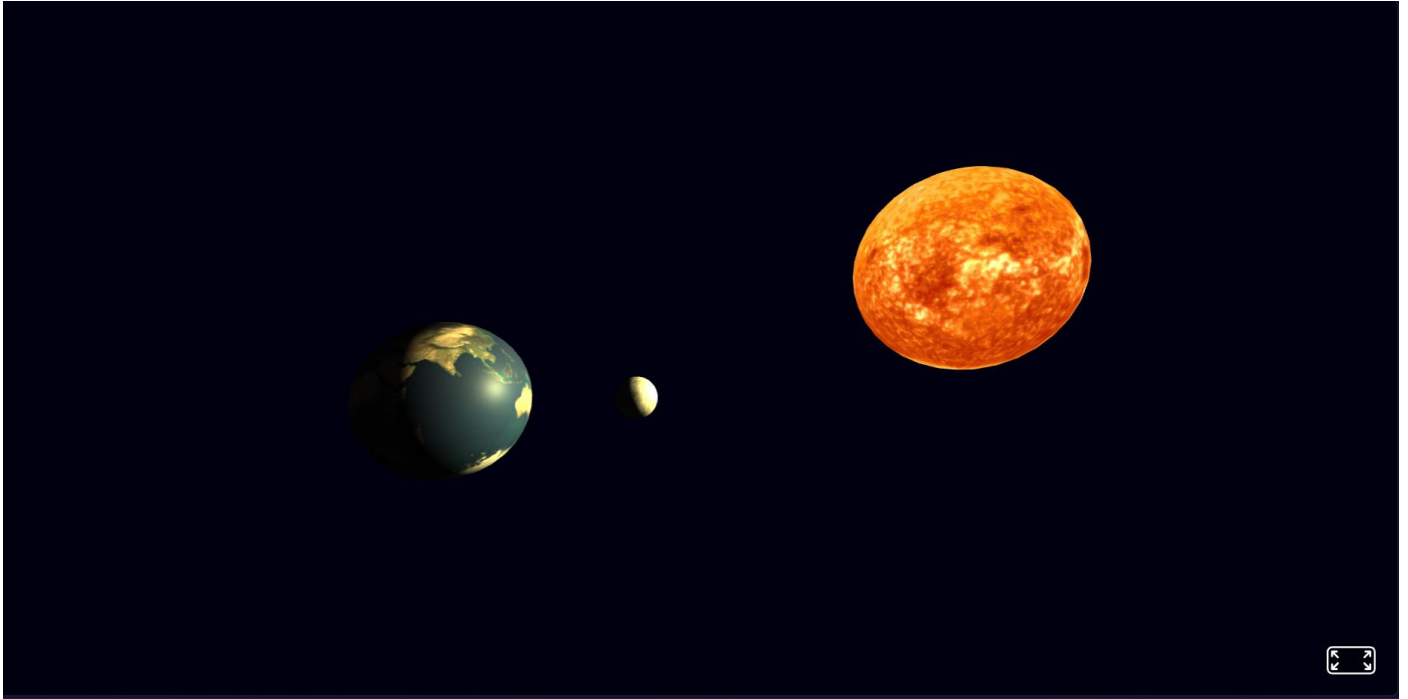


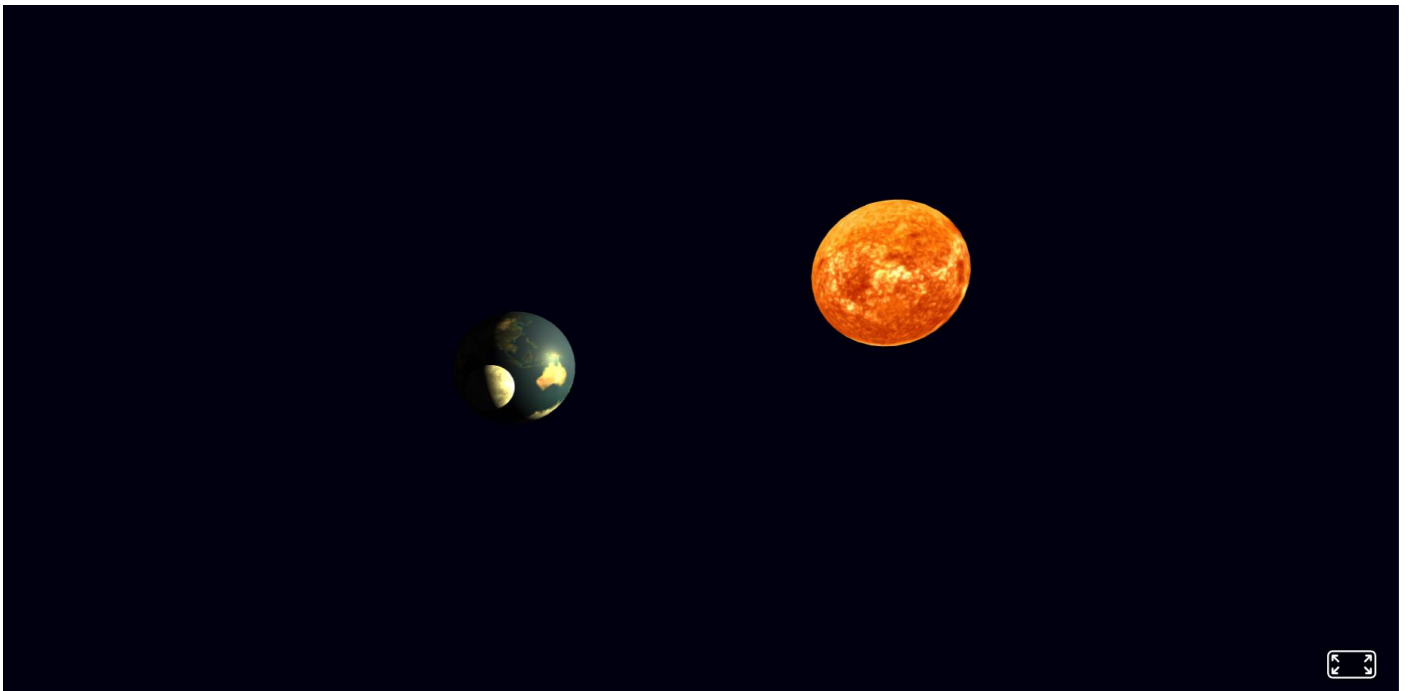
ASSIGNMENT-AR/VR

Planetary Simulation of Earth&Moon

Moon at One point



Moon at another point



This Earth-moon simulation is made by a-frame. I placed a sun for lighting. Otherwise, it would be dark. The sun is a pointlight with ambientlight also applied, so the lighting would be realistic and other dark spots won't be too dark of earth and moon.

I created an a-entity called moon-orbit and placed it at the exact same position as the Earth. I applied a rotation animation to this parent entity. Because this entity rotates, anything inside it will rotate around that center point. Basically a parent entity for just the rotation

Inside that orbit entity, I placed the Moon's sphere. Because it is a child of the rotating orbit entity, it follows a circular path, creating a Revolution like orbiting.

Then, to make the Moon spin on its own axis while it is moving, I added a second animation component directly to the Moon sphere itself.

Parent entity controls the orbital speed and child entity controls the rotation

Also, I put the same logic of assignment 2 which is the Earth tilting 23 degree and Earth's rotation.